

Fitness cost of drug resistance in *Mycobacterium tuberculosis*

S. Gagneux

Division of Mycobacterial Research, MRC National Institute for Medical Research, Mill Hill, London, UK

Abstract

Multidrug-resistant (MDR) - and extensively drug-resistant (XDR) - forms of tuberculosis are growing public health problems. Mathematical models predict that the future of the MDR and XDR tuberculosis epidemics depends in part on the competitive fitness of drug-resistant strains. Here, recent experimental and molecular epidemiological data that illustrate how heterogeneity among drug-resistant strains of *Mycobacterium tuberculosis* can influence the relative fitness and transmission of this pathogen are reviewed.

Keywords: Antibiotic, competition, genotype, mutation, mycobacteria, transmission

Clin Microbiol Infect 2009; **15** (Suppl. 1): 66–68

Corresponding author and reprint requests: S. Gagneux, Division of Mycobacterial Research, MRC National Institute for Medical Research, Mill Hill, London, UK
E-mail: gagneux@nimr.mrc.ac.uk

Drug resistance has become a problem of global proportions, with major consequences for public health and the world economy [1]. In *Mycobacterium tuberculosis*, the emergence of multidrug-resistant (MDR) and extensively drug-resistant (XDR) strains is threatening to make one of the most important infectious diseases untreatable [2]. MDR strains of *M. tuberculosis* are defined as resistant to at least isoniazid and rifampicin, the two most potent first-line tuberculosis drugs. XDR strains of *M. tuberculosis* are MDR strains with additional resistance to fluoroquinolones and to at least one of the three injectable second-line tuberculosis drugs. Recent reports of XDR tuberculosis from KwaZulu-Natal in South Africa have illustrated the dramatic negative impact that drug-resistant tuberculosis can have on patient survival, particularly in areas with high rates of human immunodeficiency virus co-infection [3].

Understanding the ecology of drug-resistant pathogens is essential for devising rational programmes to preserve the effective lifespan of antimicrobial agents and to abrogate epidemics of drug-resistant organisms. Studies in model organisms have shown that drug resistance is often associated with a fitness cost (i.e. reduced growth rate in the absence of the drug) [4]. However, particular drug resistance-conferring mutations can have a variable impact on bacterial fitness, which varies according to the specific resistance-conferring mutation and strain genetic background. Furthermore, compensatory evolution can reduce initial fitness defects caused by particular drug resistance-conferring mutations [5]. In

M. tuberculosis, mathematical models predict that the future of MDR tuberculosis epidemics will be influenced by strain fitness [6,7]. However, empirical data are lacking. We have used a combination of *in vitro* fitness assays and molecular epidemiological studies to investigate the effect of bacterial genetics on the competitive fitness and relative transmission of drug-resistant *M. tuberculosis*.

To study the influence of different rifampicin resistance-conferring mutations on the fitness of rifampicin-resistant strains of *M. tuberculosis*, we selected for spontaneous *rpoB* mutants *in vitro* using two different pan-susceptible parental strains [8]. We performed competitive fitness experiments *in vitro*, and found that in these laboratory-derived mutants, rifampicin resistance was universally associated with a competitive fitness cost (Fig. 1). However, the magnitude of this defect was determined by the specific resistance mutation and strain genetic background. For example, whereas the *rpoB* S531L mutant had higher competitive ability than other mutants in both strain backgrounds, the *rpoB* H526D mutation had a different fitness effect in the two strain backgrounds. Interestingly, when we compared clinical strains of *M. tuberculosis* that had acquired rifampicin resistance during patient treatment, we found that four out of five clinical strains with the *rpoB* S531L mutations had no fitness cost as compared to the initial rifampicin-susceptible isolate recovered from the same patient (Fig. 2). This suggests that either the *rpoB* S531L change caused no fitness defect in these particular strain backgrounds, or that the low initial fitness costs associated with the *rpoB* S531L mutation were mitigated by compensatory evolution during prolonged patient treatment. Importantly, *rpoB* S531L, the mutation associated with the lowest fitness cost in the laboratory mutants and no fitness cost in some clinical strains, was also the most

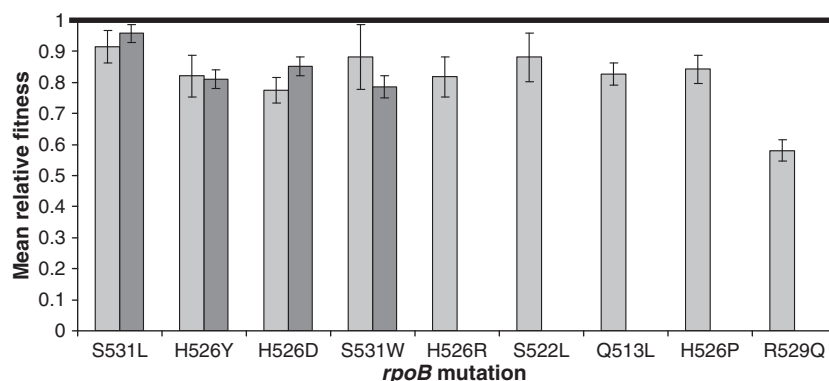


Fig. 1. Relative competitive fitness of laboratory-derived rifampicin-resistant mutants of *Mycobacterium tuberculosis*. The relative fitness of the rifampicin-susceptible ancestor is defined as 1 (black line). All rifampicin-resistant mutants had a statistically significant fitness cost as compared to the rifampicin-susceptible ancestor strain (error bars indicate 95% CIs). This cost was less in *rpoB* S531L mutants than in other *rpoB* mutants, irrespective of the strain background. Light grey bars, CDC1551 mutants; dark grey bars, T85 mutants.

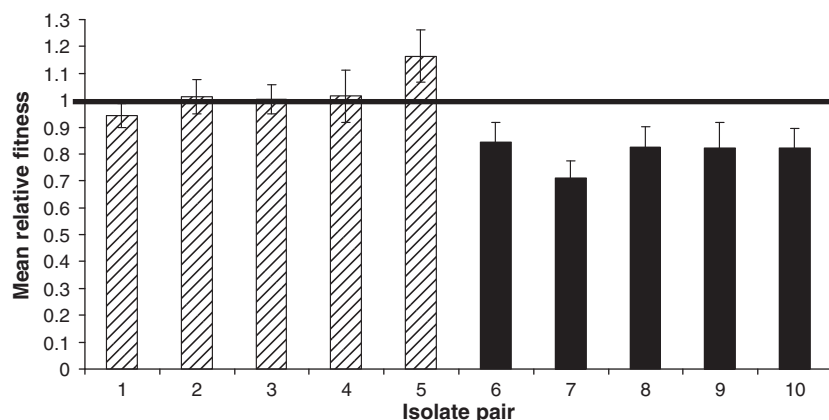


Fig. 2. Relative competitive fitness of clinically derived rifampicin-resistant mutants of *Mycobacterium tuberculosis*. Four of the five mutants with the *rpoB* S531L mutation (hatched bars) had no fitness cost as compared with their rifampicin-susceptible ancestors. All mutants with other *rpoB* mutations (black bars) had significant fitness defects (error bars indicate 95% CIs).

frequent rifampicin resistance mutation among clinical isolates globally. By contrast, the *rpoB* R529Q mutant, which carried the highest fitness cost of all mutants, has never been observed in clinical settings.

In a second study, we investigated the role of different isoniazid resistance-conferring mutations on the transmission of isoniazid-resistant *M. tuberculosis* [9]. Isoniazid is a prodrug that needs to be activated to its bioactive form by the mycobacterial katalase-peroxidase encoded by *katG* [10]. Deletions or missense mutations in *katG* generally lead to isoniazid resistance by disabling drug activation. Because *katG* helps to protect the mycobacterial cell against host-mediated oxidative stress, strains with disrupted *katG* suffer virulence defects in animal models. However, one particular *katG* mutation, S315T, is associated with a high degree of isoniazid resistance but retains enzyme activity and virulence in mice

[11]. Using a population-based molecular epidemiological approach, we measured the relative transmission of isoniazid-resistant strains of *M. tuberculosis* harbouring different *katG* mutations in San Francisco during a 9-year period [9]. We found that only strains with the *katG* S315T mutation or mutations outside of *katG* were associated with successful transmission. In contrast, strains with isoniazid resistance-conferring mutations likely to abrogate *katG* activity did not cause one secondary case during the whole study period. These results were consistent with the experimental evidence reviewed above, and further demonstrate that in *M. tuberculosis*, different drug resistance-conferring mutations can have variable effects on strain fitness and transmission.

M. tuberculosis has a phylogeographic population structure, with different strain lineages being associated with different geographical regions [12,13]. When we compared the distri-

Table 1. Association between three *Mycobacterium tuberculosis* strain lineages and three classes of isoniazid resistance-conferring mutations

Mutation	Beijing lineage (N = 36) n (%)	Ancestral lineage (N = 45) n (%)	Euro-American lineage (N = 61) n (%)	OR (95% CI) ^a	p
<i>katG</i> mutations other than S315T	15 (41.7)	2 (4.4)	10 (16.4)	5.6 (2.1–15.1)	<0.001
<i>katG</i> S315T	13 (36.1)	15 (33.3)	31 (50.8)	2.0 (0.94–4.1)	0.0517
<i>inhA</i> promoter –15; c-t	5 (13.9)	21 (46.7)	13 (21.3)	3.8 (1.6–9.0)	<0.001
No/other mutation	3 (8.3)	7 (15.6)	7 (11.5)		

^aFor each lineage, the χ^2 test was used to compare the proportion of the most frequent mutation with its proportion in the two other lineages combined.

bution of strain lineages in the isoniazid-resistant strains from San Francisco, we found statistically significant associations among the three strain lineages that dominate in San Francisco and three groups of isoniazid resistance-conferring mutations (Table 1). The 'Beijing' lineage was associated with *katG* mutations other than S315T, which were likely to abrogate *katG* activity. The *inhA* promoter mutation, also associated with isoniazid resistance, occurred significantly more frequently among the 'ancestral' lineage, and the 'Euro-American' lineage was associated with the *katG* S315T mutation, which maintains *katG* activity. These associations suggest that genetic differences between the strain lineages could influence their propensity to have different isoniazid resistance-conferring mutations. In particular, the association of the 'Beijing' strains with 'costly' *katG* mutations is intriguing, as it suggests that this strain lineage might be less susceptible to host-mediated oxidative stress. Alternatively, 'Beijing' strains might be better adapted to compensate for the loss, or reduced activity, of *katG* in the context of isoniazid resistance. As a consequence of either of these scenarios, 'Beijing' strains with low-cost isoniazid resistance-conferring mutations such as *katG* S315T may be particularly fit. In support of this possibility, 'Beijing' strains have been associated with resistance to isoniazid in several geographical areas [14].

Taken together, the data from these two studies show that there is a strong selection for drug resistance-conferring mutations associated with low fitness cost in tuberculosis patients, and that bacterial genetics can impact on the fitness and transmissibility of drug-resistant *M. tuberculosis*. Our find-

ings have implications for predicting the future of the MDR and XDR tuberculosis epidemic.

Transparency Declaration

The author declares no conflicts of interest.

References

- Palumbi SR. Humans as the world's greatest evolutionary force. *Science* 2001; 293: 1786–1790.
- Raviglione MC, Smith IM. XDR tuberculosis—implications for global public health. *N Engl J Med* 2007; 356: 656–659.
- Gandhi NR, Moll A, Sturm AW, et al. Extensively drug-resistant tuberculosis as a cause of death in patients co-infected with tuberculosis and HIV in a rural area of South Africa. *Lancet* 2006; 368: 1575–1580.
- Andersson DI, Levin BR. The biological cost of antibiotic resistance. *Curr Opin Microbiol* 1999; 2: 489–493.
- Maisnier-Patin S, Andersson DI. Adaptation to the deleterious effects of antimicrobial drug resistance mutations by compensatory evolution. *Res Microbiol* 2004; 155: 360–369.
- Blower SM, Chou T. Modeling the emergence of the 'hot zones': tuberculosis and the amplification dynamics of drug resistance. *Nat Med* 2004; 10: 1111–1116.
- Cohen T, Murray M. Modeling epidemics of multidrug-resistant *M. tuberculosis* of heterogeneous fitness. *Nat Med* 2004; 10: 1117–1121.
- Gagneux S, Long CD, Small PM, Van T, Schoolnik GK, Bohannon BJ. The competitive cost of antibiotic resistance in *Mycobacterium tuberculosis*. *Science* 2006; 312: 1944–1946.
- Gagneux S, Burgos MV, DeRiemer K et al. Impact of bacterial genetics on the transmission of isoniazid-resistant *Mycobacterium tuberculosis*. *PLoS Pathog* 2006; 2: e61.
- Heym B, Alzari PM, Honore N, Cole ST. Missense mutations in the catalase-peroxidase gene, *katG*, are associated with isoniazid resistance in *Mycobacterium tuberculosis*. *Mol Microbiol* 1995; 15: 235–245.
- Pym AS, Saint-Joanis B, Cole ST. Effect of *katG* Mutations on the Virulence of *Mycobacterium tuberculosis* and the Implication for Transmission in Humans. *Infect Immun* 2002; 70: 4955–4960.
- Gagneux S, Deriemer K, Van T et al. Variable host-pathogen compatibility in *Mycobacterium tuberculosis*. *Proc Natl Acad Sci U S A* 2006; 103: 2869–2873.
- Gagneux S, Small PM. Global phylogeography of *Mycobacterium tuberculosis* and implications for tuberculosis product development. *Lancet Infect Dis* 2007; 7: 328–337.
- Bifani PJ, Mathema B, Kurepina NE, Kreiswirth BN. Global dissemination of the *Mycobacterium tuberculosis* W-Beijing family strains. *Trends Microbiol* 2002; 10: 45–52.